

Ujjwal Basumatary

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Education

Indian Institute of Science, Bangalore

Bachelor and Master of Science, Major in Physics

Nov 2020 – Apr 2025

Physics: 9.6/10 (Overall: 9.3/10)

Publications

- [2410.22702](#) *Phys. Rev. D* 111, L041306
Beyond Hawking evaporation of black holes formed by dark matter in compact stars, Ujjwal Basumatary, Nirmal Raj and Anupam Ray.

Preprints

- [2603.26090](#)
Cosmological Correlators Using Tensor Networks, Ujjwal Basumatary, Aninda Sinha and Xinan Zhou.
- [2512.02706](#)
Lectures on Quantum Field Theory on a Quantum Computer, Aninda Sinha and Ujjwal Basumatary.

Employment

Raman Research Institute, Bangalore

Project intern

Aug 2025 – Jun 2026

National Quantum Mission

Indian Institute of Science, Bangalore

Project Associate - 1

Jun 2025 – Jul 2025

Teaching assistantships

HE381: Quantum Field Theory and Quantum Computing

Instructor: Prof. Aninda Sinha

IISc, Bangalore

CHEP

- Conducted tutorials and designed/graded assignments, with demonstrations ranging from basic quantum algorithms to advanced Matrix Product States (MPS) algorithms.
- The implementation details of the quantum and MPS algorithms are in [this](#) GitHub repository.

Fellowships and Awards

S. N. Bhatt Memorial Fellowship

International Center for Theoretical Sciences (ICTS)

May 2023 – Jul 2023

Bangalore

Kishore Vaiyanik Protsahan Yojana (KVPY) Fellowship
Department of Science and Technology (DST)

Nov 2020 – Apr 2025
India

National Talent Search Exam (NTSE) Scholarship
National Council of Educational Research & Training (NCERT)

Apr 2018 – Mar 2020
India

Skills

- **Programming:** Julia, Python, Mathematica, C, C++ and Haskell.
- **Markup:** Proficient in $\text{\LaTeX}$, working knowledge of HTML and CSS

Advanced Graduate Coursework

Non-eq. Many-Body Dynamics	10/10	AdS ₃ /CFT ₂	9/10
Standard Model	10/10	Adv. Mathematical Physics	10/10
QFT - 2	9/10	Cosmology for Theorists	9/10
Statistical Field Theory	9/10	QFT - 1	10/10
General Relativity	10/10	Nuclear and Particle Physics	10/10